

Inductive bias meta-learning with generative models

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Goal of research

Goal

Extraction of inductive bias of differentiable learning algorithms.

Problem

The existing solution assumes prior knowledge of the data distribution or requires a fixed dataset.

Solution

A meta-learning algorithm that trains a generator to produce datasets that are easy to generalize to the target model and that reflect its inductive bias.

Extracting inductive bias via generating easy-to-generalize datasets

Previous work^a

Meta-learning a function that assigns labels to samples from a fixed dataset or data distribution.

Our extension

Meta-learning a generative model to produce entire synthetic datasets.

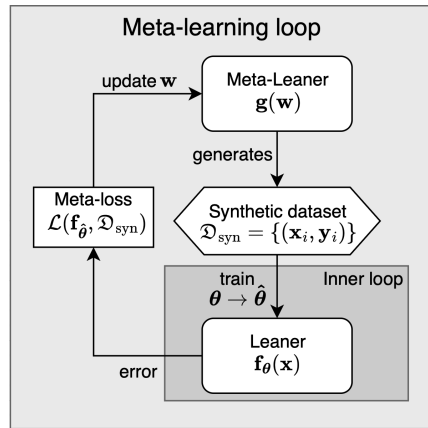


Figure: Meta-learning scheme.

^aWilliam Dorrell, Maria Yuffa, and Peter Latham. *Meta-Learning the Inductive Biases of Simple Neural Circuits*. 2023. arXiv: 2211.13544 [q-bio.NC]. URL: <https://arxiv.org/abs/2211.13544>.

Problem statement

Given a target model $\mathbf{f}_{\boldsymbol{\theta}} : \mathbb{X} \rightarrow \mathbb{Y}$ with loss function $\ell(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y})$, the generator $\mathbf{g}(\mathbf{w})$ defines a parametric family of conditional distributions $p(\mathbf{x} \mid \mathbf{y}, \mathbf{w})$ over inputs $\mathbf{x}$ given labels $\mathbf{y}$.

Required to find data distribution function p^* , corresponding to parameters $\mathbf{w}^*$, the solution to the following optimization problem:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \mathbb{E}_{\mathcal{D}_{\text{syn}} \sim p(\mathbf{x}|\mathbf{y}, \mathbf{w})} [\mathcal{L}(\mathbf{f}_{\hat{\boldsymbol{\theta}}}, \mathcal{D}_{\text{syn}})] \quad \text{s.t.} \quad \mathcal{D}_{\text{syn}} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}} \in \mathcal{U}$$

where $\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{train}}} \ell(\mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y})$, $\mathcal{L}$ is meta loss function and $\mathcal{U}$ is the set of datasets consisting of c classes with n samples per class.

Meta loss function

The meta-loss function $\mathcal{L}$ combines prediction error $\mathcal{L}_g$ and regularization $\mathcal{R}$:

$$\mathcal{L} = \mathcal{L}_g + \lambda \mathcal{R}$$

Prediction error $\mathcal{L}_g$ (MSE or cross-entropy):

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{test}}} \sum_{i=1}^m y_i \log([\mathbf{f}_{\hat{\theta}}(\mathbf{x})]_i)$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{test}}} \|\mathbf{y} - \mathbf{f}_{\hat{\theta}}(\mathbf{x})\|^2$$

Regularization $\mathcal{R}$:

$$\mathcal{R}_{\text{dist}} = -\frac{1}{C} \sum_{k=1}^C \frac{1}{n_k(n_k - 1)} \sum_{\substack{i, j \in \mathcal{I}_k \\ i \neq j}} \|\mathbf{x}_i - \mathbf{x}_j\|_2^2, \quad \mathcal{R}_{\text{svd}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{r_k} \sum_{i=1}^{r_k} (\max(\tau - \sigma_i^{(k)}, 0))^2$$

Algorithm 1 Meta-learning for inductive bias extraction

- 1: Initialize generator $g(\mathbf{w})$ with random parameters $\mathbf{w}$
 - 2: **while** step $<$ total steps **do**
 - 3: Generate synthetic dataset $\mathcal{D}_{\text{syn}} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$ with C classes and N samples per class
 - 4: Split $\mathcal{D}_{\text{syn}}$ into training set $\mathcal{D}_{\text{train}}$ and test set $\mathcal{D}_{\text{test}}$
 - 5: Re-initialize target model $\mathbf{f}_{\theta}$ from scratch
 - 6: Train $\mathbf{f}_{\theta}$ on $\mathcal{D}_{\text{train}}$ in the inner loop $\rightarrow$ trained model $\mathbf{f}_{\hat{\theta}}$
 - 7: Compute meta-loss: $\mathcal{L}_{\text{meta}} = \mathcal{L}_g(\mathbf{f}_{\hat{\theta}}, \mathcal{D}_{\text{test}}) + \lambda \mathcal{R}(\mathcal{D}_{\text{syn}})$
 - 8: Take $\mathbf{w}$ gradient step on meta-loss
 - 9: **end while**
-

CNN and 1x4 images

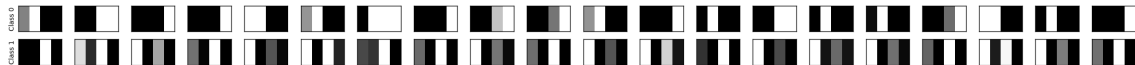


Figure: Obtained dataset for average intra-class distance regularization.

Training	Test	Accuracy (%)
Original	Original	98.78 ± 8.82
Original	Swapped	98.78 ± 8.82
Swapped	Swapped	99.08 ± 7.25
Swapped	Original	99.08 ± 7.25

CNN Architecture: Conv1d with kernel 1×2 with stride 1x2.

Meta-loss function: $\mathcal{L} = \mathcal{L}_{MSE} + 0.5 \cdot \mathcal{R}_{dist}$

Table: Accuracy on original and swapped data.
The accuracy of the model trained on the original dataset does not decrease on the swapped test set, confirming locality of the obtained data.

CNN and 4x4 images: locality

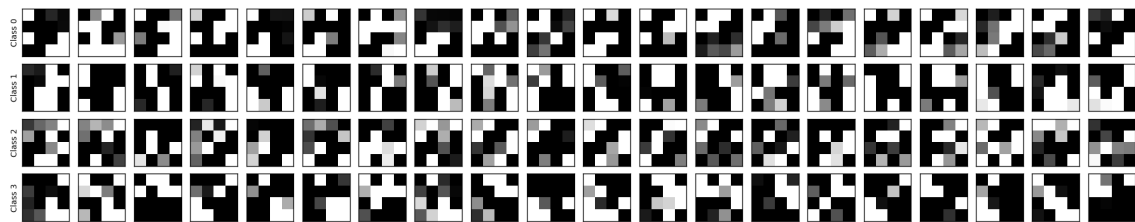


Figure: Obtained dataset for cross-entropy and average intra-class distance regularization.

CNN Architecture: Conv2d with kernel 2×2 , with stride 2×2 .

Meta-loss function: $\mathcal{L}_{CE} / \mathcal{L}_{MSE} + \mathcal{R}_{dist} / \mathcal{R}_{SVD}$ (different combinations).

CNN and 4x4 images: locality

Setup	Original	Seed 42	Seed 123	Seed 456	Seed 789
CE+Dist	98.87 ± 5.33	98.87 ± 5.33	98.87 ± 5.33	98.87 ± 5.33	98.87 ± 5.33
MSE+Dist	99.03 ± 5.22	99.03 ± 5.22	99.03 ± 5.22	99.03 ± 5.22	99.03 ± 5.22
CE+SVD	99.33 ± 4.62	99.33 ± 4.62	99.33 ± 4.62	99.33 ± 4.62	99.33 ± 4.62
MSE+SVD	98.96 ± 5.46	98.96 ± 5.46	98.96 ± 5.46	98.96 ± 5.46	98.96 ± 5.46
MLP	62.24 ± 14.73	78.76 ± 13.02	77.41 ± 12.97	77.61 ± 12.97	76.05 ± 13.68

Table: Accuracy comparison on original and randomly shuffled data for all meta-loss configurations. The CNN trained on the original dataset preserves its accuracy on shuffled test data, confirming the locality of the generated data. In contrast, the MLP achieves notably lower performance.

CNN and 4x4 images: translation invariance

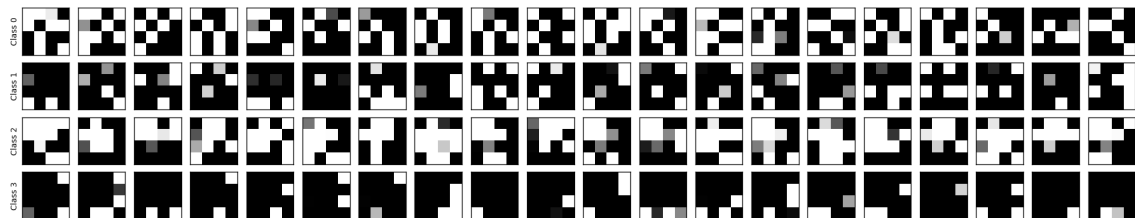


Figure: Obtained dataset.

Shift type	Accuracy (%)
Original	98.78 ± 8.82
Shift horizontal (1px)	95.23 ± 10.12
Shift vertical (1px)	94.56 ± 11.34
Shift horizontal+vertical (1px)	89.12 ± 14.56

Table: Accuracy comparison on original and shifted images. Only a slight drop in accuracy is observed under translation, indicating translation invariance.

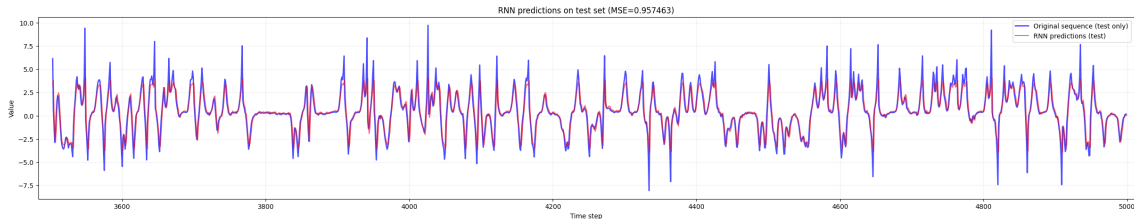


Figure: RNN prediction on test set.

Training	Test	MSE
Original	Original	1.13 ± 0.04
Original	Shuffled	7.21 ± 0.26
Shuffled	Shuffled	5.973 ± 0.004

Table: MSE comparison on original and shuffled data. The model performs notably worse on randomly shuffled data, highlighting the importance of sequential order.

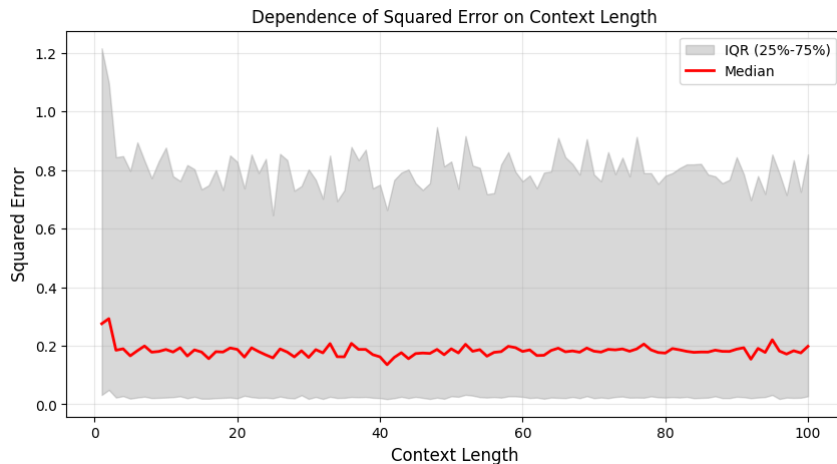


Figure: Dependence of Squared Error on context length. The median error decreases slightly for short contexts and then remains constant, confirming the presence of a sequential structure.

Conclusion

- 1) We consider the problem of extracting the inductive bias of differentiable learning algorithms without prior knowledge of the input data distribution.
- 2) We propose a generative meta-learning framework where a generative model is trained to produce synthetic easy-to-generalize datasets that reflect the target model's inductive bias.
- 3) Experiments on CNNs and RNNs demonstrate that the generated datasets reflect the well-known inductive biases of these architectures: locality and translation invariance for CNNs, and sequential structure for RNNs.
- 4) Our method successfully extracts inductive bias without requiring any external data, offering a tool for analyzing different model architectures.